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AMPLITUDE MODULATION [AM]

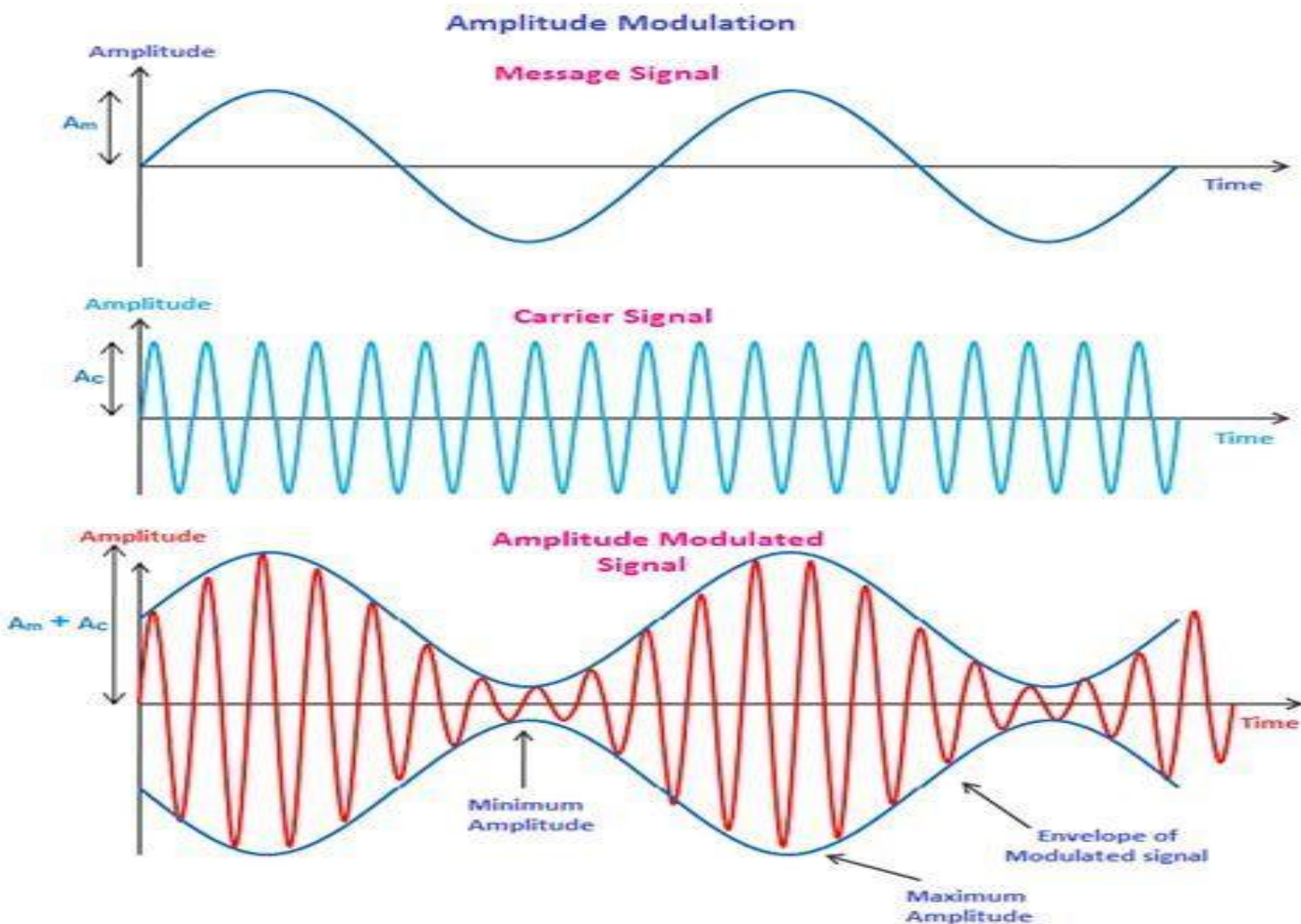
Definition:

Amplitude Modulation is a modulation technique in which the amplitude of a carrier signal is varied according to the information (message) signal while the carrier frequency remains constant.

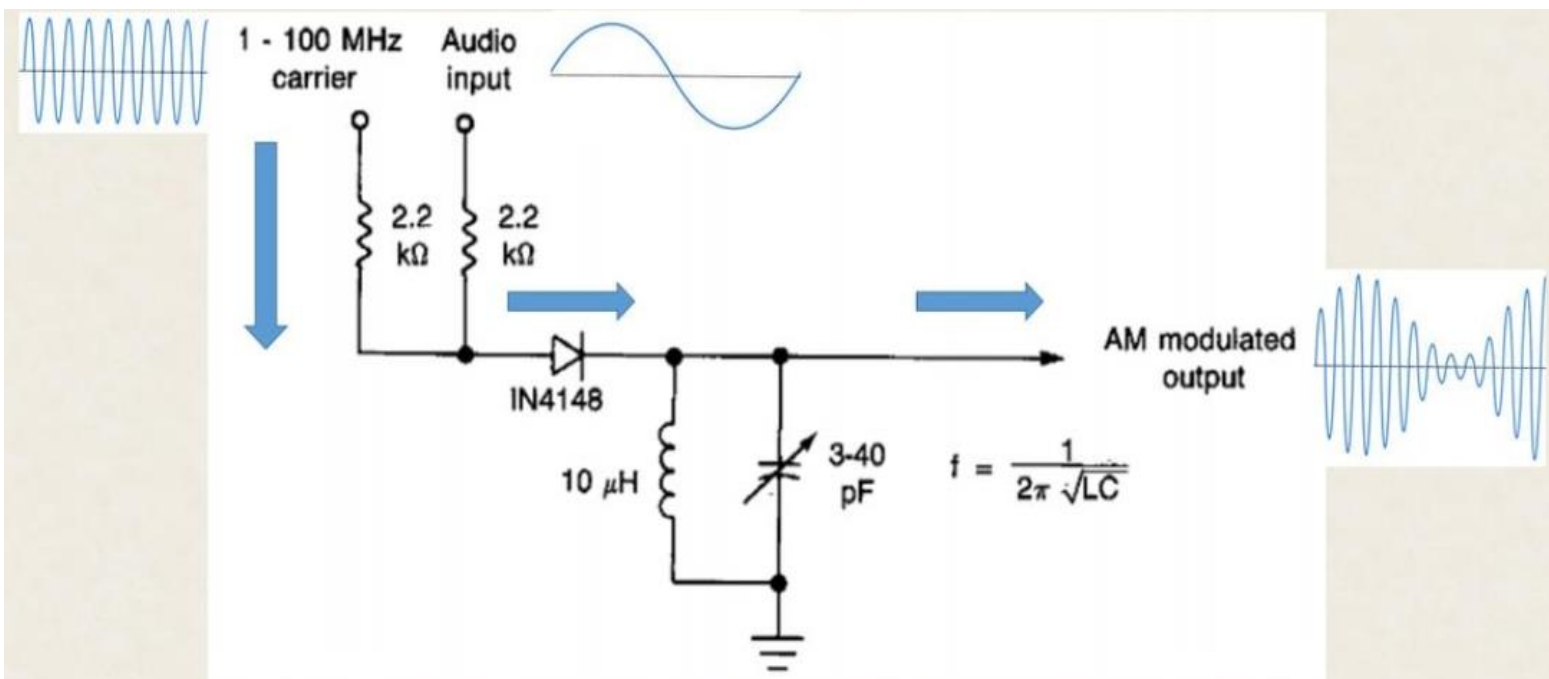
Applications:

- AM Radio Broadcasting
- Aircraft Communication Systems
- Long-Distance Radio Communication

Waveform:



Amplitude Modulation Circuit Diagram:



AM (Amplitude Modulation) Equations

1) Message Signal

$$m(t) = A_m \cos(2\pi f_m t)$$

A_m : Amplitude of the message signal
 f_m : Frequency of the message signal

2) Carrier Signal

$$c(t) = A_c \cos(2\pi f_c t)$$

A_c : Amplitude of the carrier signal
 f_c : Frequency of the carrier signal
 $(f_c \gg f_m)$

3) AM Wave (Time Domain)

$$s(t) = A_c [1 + \mu m(t)] \cos(2\pi f_c t)$$

$s(t)$: AM modulated signal

μ : Modulation index ($0 \leq \mu \leq 1$)

$m(t)$: Normalized message signal, where $|m(t)| \leq 1$

4) Modulation Index

$$\mu = \frac{A_m}{A_c}$$

A_m : Peak amplitude of message signal

A_c : Amplitude of carrier signal

5) AM Wave (Expanded Form)

$$\begin{aligned} s(t) &= A_c \cos(2\pi f_c t) + \mu A_c m(t) \cos(2\pi f_c t) \\ &= A_c \cos(2\pi f_c t) + \frac{\mu A_c A_m}{2} [\cos 2\pi(f_c + f_m)t + \cos 2\pi(f_c - f_m)t] \end{aligned}$$

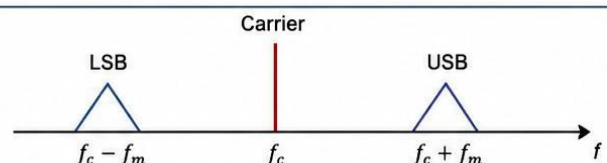
This shows that AM wave contains three frequency components:

- Carrier component at f_c
- Upper sideband (USB) at $f_c + f_m$
- Lower sideband (LSB) at $f_c - f_m$

6) Bandwidth of AM Signal

$$BW = 2f_m$$

Where f_m is the highest frequency of the message signal.



FREQUENCY MODULATION [FM]

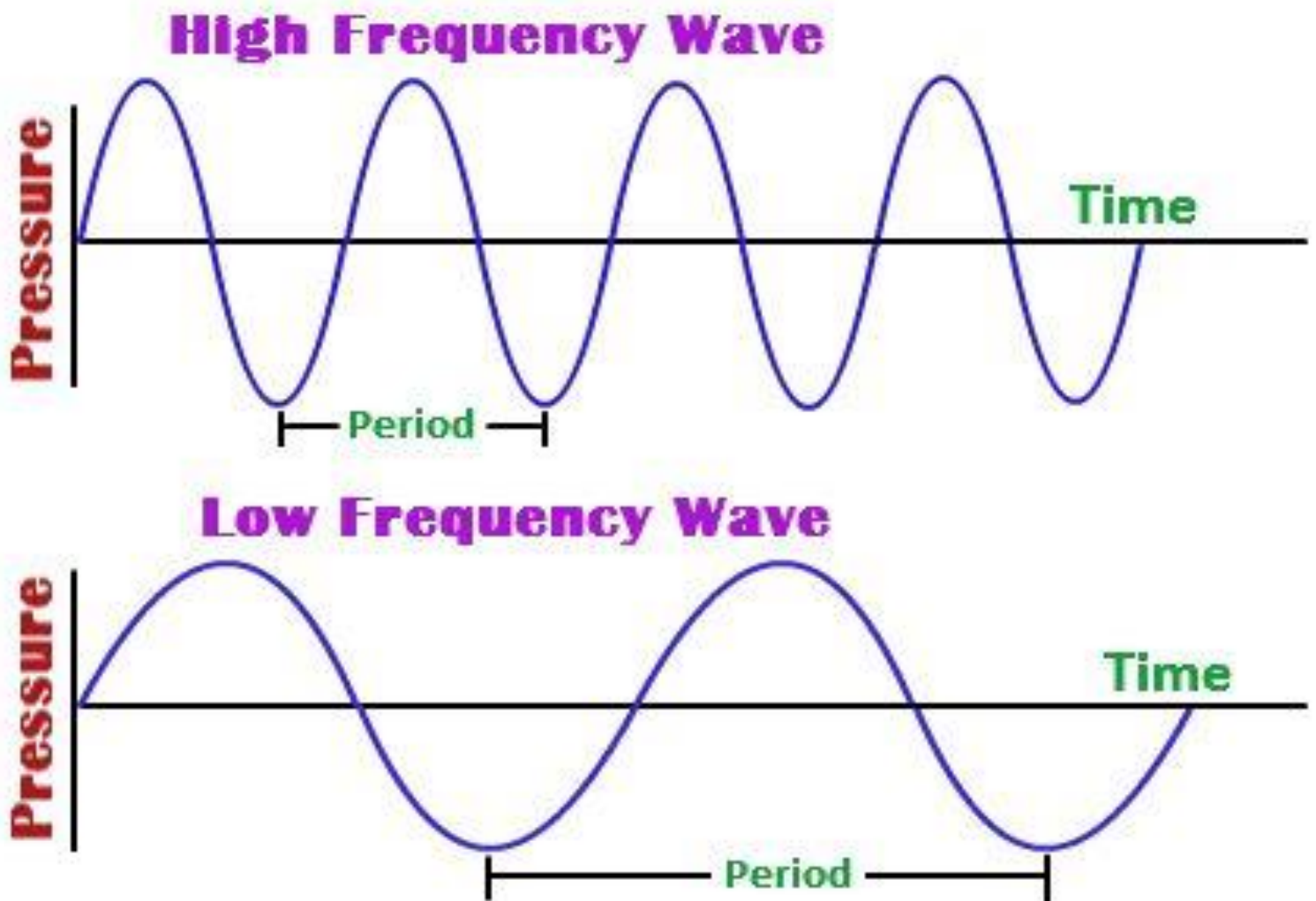
Definition:

Frequency Modulation is a modulation technique in which the frequency of the carrier signal is varied according to the message signal while the amplitude remains constant.

Applications:

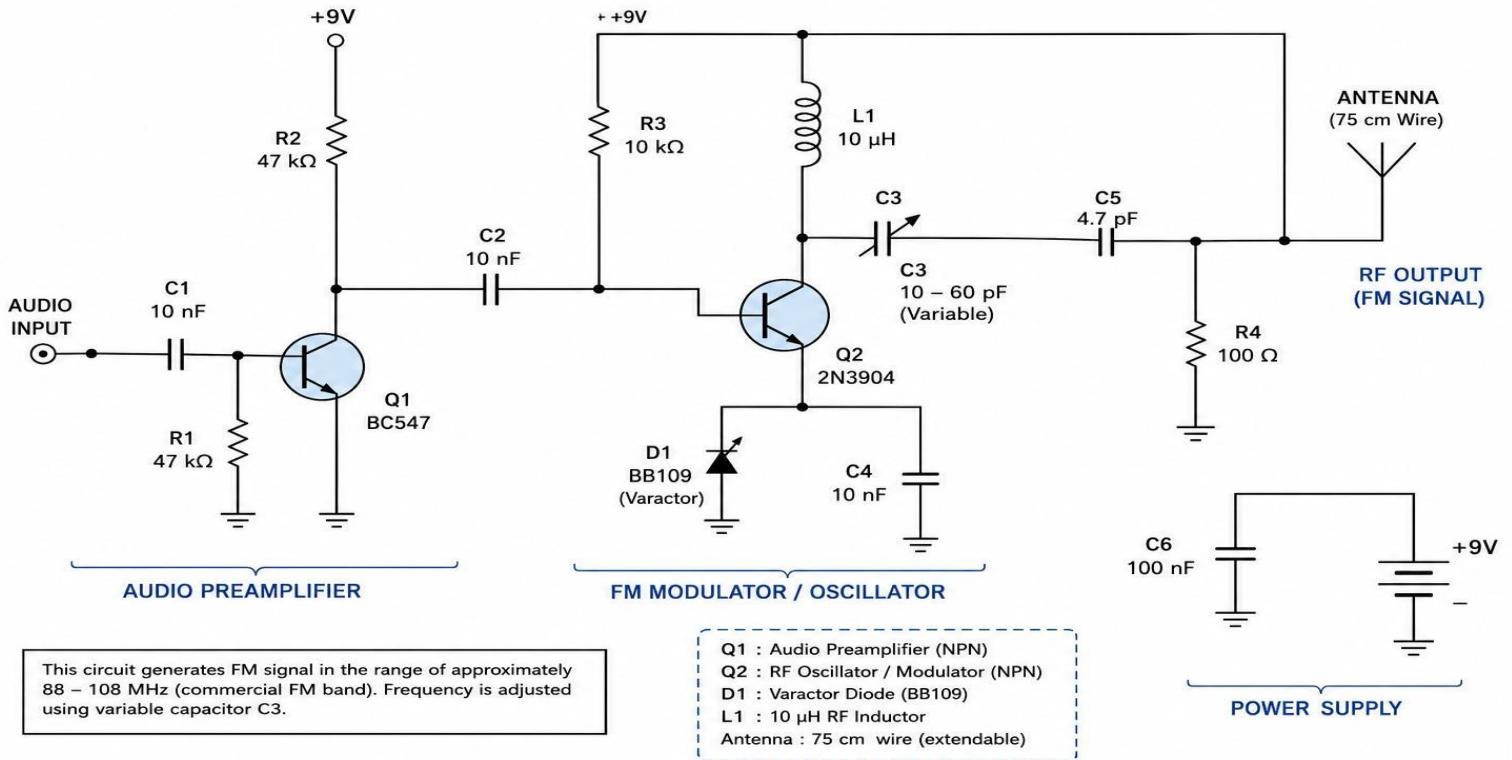
- FM Radio Broadcasting
- Wireless Communications Systems
- Television Sound Transmission

Waveform:



Frequency Modulation Circuit Diagram:

FM TRANSMITTER CIRCUIT



FM (Frequency Modulation) Equations

1) Message Signal

$$m(t) = A_m \cos(2\pi f_m t)$$

A_m : Amplitude of the message signal
 f_m : Frequency of the message signal

2) Carrier Signal

$$c(t) = A_c \cos(2\pi f_c t)$$

A_c : Amplitude of the carrier signal
 f_c : Carrier frequency
($f_c \gg f_m$)

3) Instantaneous Frequency

$$f_i(t) = f_c + k_f m(t)$$

$f_i(t)$: Instantaneous frequency
 k_f : Frequency sensitivity constant (Hz / Volt)

4) FM Wave (Time Domain)

$$s(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau\right)$$

$s(t)$: FM modulated signal
 A_c : Amplitude of the carrier signal (constant)
 k_f : Frequency sensitivity constant
 $\int_0^t m(\tau) d\tau$: Integral of the message signal

6) Modulation Index (FM Index)

$$\beta = \frac{\Delta f}{f_m}$$

β : Modulation index (dimensionless)
 Δf : Maximum frequency deviation
 f_m : Highest frequency of message signal

7) Carson's Rule (Bandwidth of FM Signal)

$$BW \approx 2(\Delta f + f_m) = 2(\beta + 1) f_m$$

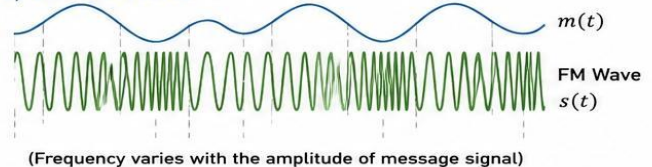
BW : Approximate bandwidth of FM signal
 Δf : Maximum frequency deviation
 f_m : Highest frequency of message signal
 β : Modulation index

5) Frequency Deviation

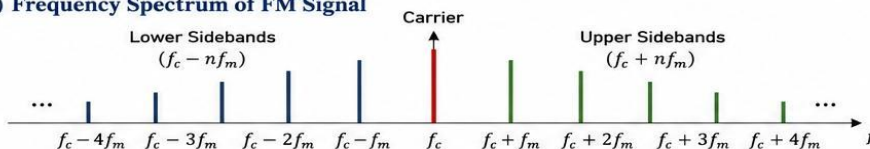
$$\Delta f = k_f A_m$$

Δf : Maximum frequency deviation (Hz)
 A_m : Peak amplitude of $m(t)$

8) FM Waveform



9) Frequency Spectrum of FM Signal



Notes:

- In FM, the amplitude remains constant, while the frequency varies.
- The greater the modulating signal amplitude or k_f , the larger the frequency deviation.
- When $\beta \gg 1$, FM is said to be wideband FM.

FILTERS

Definition:

Filter:

A filter is an electronic circuit that allows certain frequencies to pass while attenuating or blocking others.

Ideal Filter:

An ideal filter is a theoretical model that does not exist in real life, but is used for analysis.

Applications:

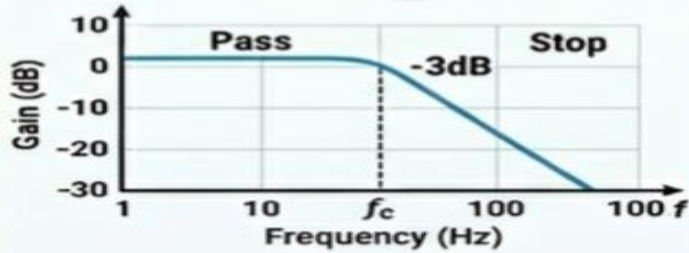
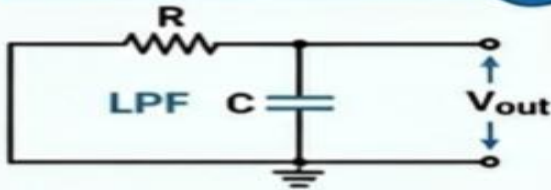
- Radio Receivers and Transmitters
- Audio Processing Systems
- Communication Systems
- Power Supplies

Types Of Filters:

- ① **Low-Pass Filter [LPF]:** Allows low frequencies to pass and blocks high frequencies.
- ② **High-Pass Filter [HPF]:** Allows high frequencies to pass and blocks low frequencies.
- ③ **Band-Pass Filter [BPS]:** Allows a specific range of frequencies to pass.
- ④ **Band-Stop Filter [BSF]:** Blocks a specific range of frequencies.

RC LOW-PASS FILTER (LPF)

Low-Pass Filter



Cut-off
Frequency

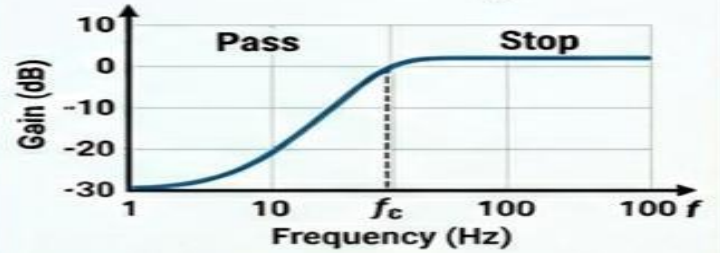
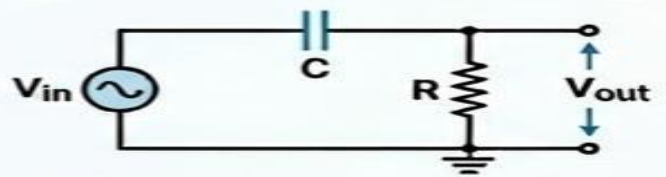
$$f_c = \frac{1}{2\pi RC}$$

Transfer
Function

$$H(s) = \frac{1}{1 + sRC}$$

RC HIGH-PASS FILTER (HPF)

High-Pass Filter



Cut-off
Frequency

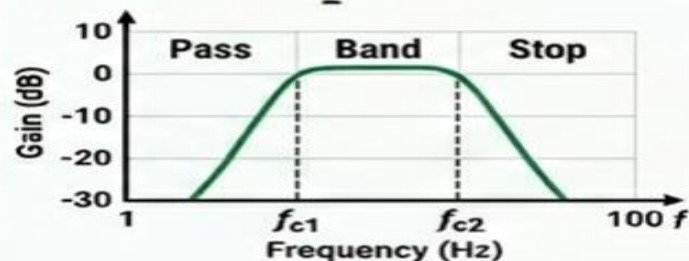
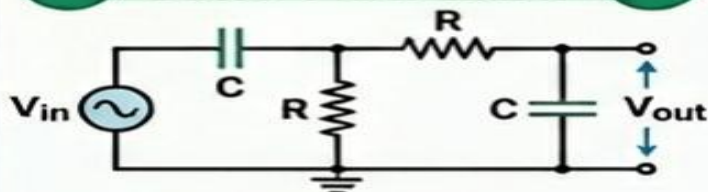
$$f_c = \frac{1}{2\pi RC}$$

Transfer
Function

$$H(s) = \frac{sRC}{1 + sRC}$$

RC BAND-PASS FILTER (BPF)

Band-Pass Filter



Center
Frequency

$$f_o = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}}$$

Bandwidth

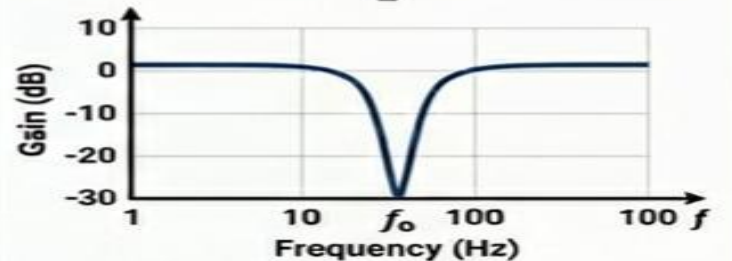
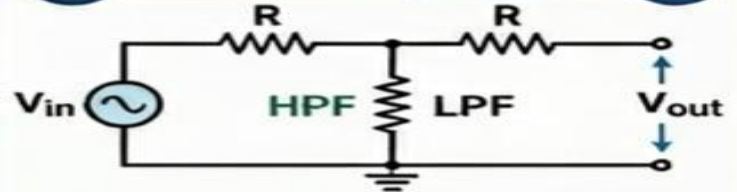
$$BW = f_{c2} - f_{c1}$$

Quality
Factor

$$Q = \frac{f_o}{BW}$$

RC BAND-STOP FILTER (BSF)

Band-Stop Filter



Notch
Frequency

$$f_o = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}}$$

Bandwidth

$$BW$$

Q Factor

$$Q$$